

Chapter 8: Nature of Matter: Elements, Compounds, and Mixtures

Q1. Which of the entities in the picture above consist of matter, and which of them do not?



Fig. 8.1: Poha

- **Answer:**
 - **Consist of Matter:** Poha, sprout salad, utensils, water, air, staircase, clothes, shoes, books, and trees. (Matter is anything that has mass and occupies space).
 - **Do NOT Consist of Matter:** Sunlight/light, heat, sound, electricity, thoughts, and emotions. (These are forms of energy or mental states that do not possess mass or volume).

Q2. How can elements be combined to form a compound?

- **Answer:** Elements combine through **chemical reactions** when two or more different elements react in fixed proportion by mass/atoms. During this chemical combination, atoms of different elements bond tightly together to form molecules of a new substance (a compound) with properties completely different from the constituent elements .

Q3. How could the discovery of a compound that absorbs carbon dioxide from the air contribute to solving environmental challenges?

- **Answer:** Carbon dioxide (CO_2) is a major greenhouse gas responsible for trapping heat and driving global warming and climate change. A compound that absorbs CO_2 from the atmosphere can be utilized in carbon-capture technologies to reduce greenhouse gas emissions, combat atmospheric pollution, and mitigate global warming.

2. Keep the Curiosity Alive (End-of-Chapter Exercises)

Q1. Consider the following reaction where two substances, A and B, combine to form a product C:



Assume that A and B cannot be broken down into simpler substances by chemical reactions. Based on this information, which of the following statements is correct?

- (i) A, B, and C are all compounds and only C has a fixed composition.
- (ii) C is a compound, and A and B have a fixed composition.
- (iii) A and B are compounds, and C has a fixed composition.
- (iv) A and B are elements, C is a compound, and has a fixed composition.

Correct Answer: (iv) A and B are elements, C is a compound, and has a fixed composition.

- **Reasoning:** Since A and B cannot be broken down into simpler substances by chemical reactions, they meet the fundamental definition of **elements**. When two elements combine chemically, they form a **compound** (C) which always has a fixed chemical composition.

Q2. Assertion & Reason:

- **Assertion:** Air is a mixture.
- **Reason:** A mixture is formed when two or more substances are mixed, without undergoing any chemical change.

(i) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.

(ii) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.

(iii) Assertion is true, but Reason is false.

(iv) Assertion is false, but Reason is true.

Correct Answer: (i) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.

- **Reasoning:** Air is composed of gases like nitrogen, oxygen, carbon dioxide, argon, and water vapour physically mixed together without reacting chemically, retaining their individual properties.

Q3. Water, a compound, has different properties compared to those of the elements oxygen and hydrogen from which it is formed.

Justify this statement.

- **Answer:**
 - **Hydrogen (H₂):** A highly inflammable gas that burns explosively.
 - **Oxygen (O₂):** A gas that actively supports combustion/burning.

- **Water (H₂O):** A liquid at room temperature that neither burns nor support combustion; instead, it is used to extinguish fires.
- **Conclusion:** This dramatic shift in chemical and physical properties proves that when elements chemically combine to form a compound, the resulting compound exhibits unique properties entirely distinct from its constituent elements.

Q4. In which of the following cases are all the examples correctly matched?

Give reasons in support of your answers.

- (i) Elements - water, nitrogen, iron, air.
- (ii) Uniform mixtures - minerals, seawater, bronze, air.
- (iii) Pure substances - carbon dioxide, iron, oxygen, sugar.
- (IV) Non-uniform mixtures - air, sand, brass, muddy water.

Correct Answer: (iii) Pure substances - carbon dioxide, iron, oxygen, sugar.

Reasons for Choice:

- **Set (iii) is CORRECT:** Pure substances consist of only one type of particle throughout. Carbon dioxide and sugar are pure compounds, while iron and oxygen are pure elements.
- **Set (i) is Incorrect:** Water is a compound and air is a mixture, not elements.
- **Set (ii) is Incorrect:** Most minerals are non-uniform natural compounds or rock aggregates.
- **Set (IV) is Incorrect:** Air and brass (an alloy) are uniform (homogeneous) mixtures, not non-uniform.

Q5. Iron reacts with moist air to form iron oxide, and magnesium burns in oxygen to form magnesium oxide.

Classify all the substances involved in the above reactions as elements, compounds, or mixtures, with justification.

- **Reaction 1:** Iron + {Moist Air} → Iron Oxide (Rust)
 - **Iron: Element.** Consists of identical iron atoms that cannot be broken down further.
 - **Moist Air: Mixture.** A physical blend of gases (nitrogen, oxygen, water vapour, etc.).
 - **Iron Oxide: Compound.** Formed by the chemical reaction of iron and oxygen in fixed chemical proportions.
- **Reaction 2:** Magnesium + Oxygen → Magnesium Oxide
 - **Magnesium: Element.** A pure metallic element.
 - **Oxygen: Element.** A pure non-metallic element existing as O₂ molecules.
 - **Magnesium Oxide: Compound.** A new chemical substance created by chemical bonding of magnesium and oxygen.

Q6. Classify the following as elements, compounds, or mixtures:

Substances: Carbon dioxide, sand, seawater, magnesium oxide, muddy water, aluminium, gold, oxygen, rust, iron sulfide, glucose, air, water, fruit juice, nitrogen, sodium chloride, sulfur, hydrogen, baking soda.

Classification Table:

Elements	Compounds	Mixtures
• Aluminium	• Carbon dioxide	• Sand
• Gold	• Magnesium oxide	• Seawater
• Oxygen	• Rust (Iron oxide)	• Muddy water
• Nitrogen	• Iron sulfide	• Air
• Sulfur	• Glucose	• Fruit juice
• Hydrogen	• Water	
	• Sodium chloride	
	• Baking soda	

Pure Substances among These:

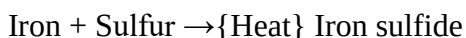
All **Elements** and **Compounds** are Pure Substances:

- *Aluminum, Gold, Oxygen, Nitrogen, Sulfur, Hydrogen, Carbon dioxide, Magnesium oxide, Rust, Iron sulfide, Glucose, Water, Sodium chloride, and Baking soda.*

Q7. What new substance is formed when a mixture of iron filings and sulfur powder is heated, and how is it different from the original mixture? Also, write the word equation for the reaction.

- **New Substance Formed: Iron sulfide (FeS).**
- **Differences from the Original Mixture:**
 1. **Appearance:** The original mixture (Sample A) shows distinct yellow sulfur powder and grey iron filings. Iron sulfide (Sample B) is a uniform, dark black mass .

2. **Magnetic Property:** Iron filings in the original mixture are attracted to a magnet. Iron sulfide is non-magnetic and is not attracted by a magnet.
 3. **Reaction with Acid:** The original mixture reacts with dilute HCl to release odorless, combustible **hydrogen gas**. Iron sulfide reacts with dilute HCl to release **hydrogen sulfide gas**, which has a foul rotten-egg smell.
- **Word Equation:**



Q8. Is it possible for a substance to be classified as both an element and a compound?

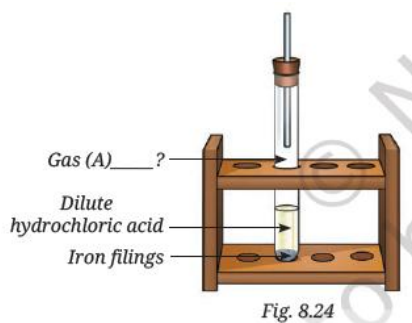
Explain why or why not.

- **Answer: No, it is impossible.**
- **Explanation:**
 - An **element** is a pure substance made up of only **one kind of atom** and cannot be broken down into simpler substances by any chemical process.
 - A **compound** is a pure substance made up of **two or more different elements** chemically combined in a fixed ratio.
 - Because these definitions are mutually exclusive, a substance cannot be both an element and a compound simultaneously.

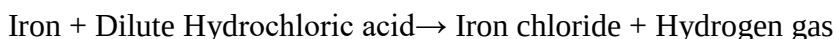
Q9. How would our daily lives be changed if water were not a compound but a mixture of hydrogen and oxygen?

- **Answer:**
 1. **No Liquid Water:** Hydrogen and oxygen would exist as separate uncombined gases at room temperature. There would be no liquid water to drink, bathe with, or support plant/animal life.
 2. **Extreme Fire Hazard:** Free hydrogen gas is explosive and combustible, while free oxygen supports burning. Any minor spark would trigger violent atmospheric explosions instead of putting out fires.
 3. **Loss of Life on Earth:** Since biological cells require liquid H₂O as a universal solvent, life as we know it could not exist.

Q10. Analyse Fig. 8.24. Identify Gas A. Also, write the word equation of the chemical reaction.



- **Identification of Gas A: Hydrogen gas (H_2).**
- **Explanation:** When iron filings react with dilute hydrochloric acid, iron replaces hydrogen to form iron chloride and liberates hydrogen gas.
- **Word Equation:**



Q11. Write the names of any two compounds made only from non-metals, and also mention two uses of each of them.

- **Compound 1: Water (H_2O)** (*Made of Non-metals: Hydrogen and Oxygen*)
 - **Use 1:** Essential for drinking, hydration, and survival of all living organisms.
 - **Use 2:** Acts as a universal solvent in household cleaning, agriculture, and industrial chemical processes.
- **Compound 2: Carbon Dioxide (CO_2)** (*Made of Non-metals: Carbon and Oxygen*)
 - **Use 1:** Used by green plants during photosynthesis to produce food and oxygen.
 - **Use 2:** Used in fire extinguishers to put out electrical/chemical fires and in carbonated beverages.

Q12. How can gold be classified as both a mineral and a metal?

- **Answer:**
 - **As a Metal:** Gold possesses fundamental physical properties of metals—it is lustrous, highly malleable, ductile, and an excellent conductor of heat and electricity.
 - **As a Mineral:** Gold occurs naturally in Earth's crust as an inorganic solid with a definite chemical structure. Because it is unreactive and found in its pure elemental form in nature, it is classified as a **native mineral**.